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Systems and methods for attachment of materials having different thermal expansion coefficients

Abstract

An apparatus for attaching two components having different coefficients of thermal expansion includes a mounting member configured to be movably coupled relative to a first component, an attachment member opposite the mounting member and configured to be attached to a second component, and at least one support post interconnecting the mounting member and the attachment member. The apparatus also includes a backer plate including at least one aperture configured to permit at least one fastener to pass therethrough and pass through the second component and through at least one aperture in the attachment member to secure the backer plate and the attachment member to the second component and to attach the second component to the first component. The diameter of the aperture of the backer plate restricts the head of the fastener from passing through the aperture of the backer plate.

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Background/Summary

TECHNICAL FIELD
(1) These teachings relate generally to jet engines and, more particularly, to attachment of aircraft engine components that have different thermal expansion coefficients.
BACKGROUND
(2) Turbine engines, and particularly gas or combustion turbine engines, are rotary engines that extract energy from a flow of combusted gases passing through the engine onto a multitude of turbine blades. Exhaust from combustion flows through a high-pressure turbine and a low-pressure turbine prior to leaving the turbine engine through an exhaust nozzle. The exhaust gas mixture passing through the exhaust nozzle is at extremely high temperatures and transfers heat to the components of the turbine engine, including the exhaust nozzle, which is typically metallic. The high-temperature environment present within the exhaust nozzle necessitates the use of materials and components that can withstand such an environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Described herein are embodiments of methods of attaching a protective liner to a metal duct of an exhaust nozzle of an aircraft engine. This description includes drawings, wherein:
(2) FIG. **1** is a schematic view of an exhaust nozzle of an aircraft engine including attachment apparatuses;
(3) FIG. **2** is a perspective side view of an attachment apparatus that can be utilized in the aircraft engine of FIG. **1** with a section exploded;
(4) FIG. **3** is a partial cross-section elevational view of the attachment apparatus of FIG. **2**;
(5) FIG. **4** is a bottom view showing an interior-facing surface of the backer plate of the attachment apparatus of FIGS. **2-3** and the heads of the fasteners that attach the backer plate to the protective liner;
(6) FIG. **5** is a partial cross-section elevational view of an attachment apparatus, that can be utilized in the aircraft engine of FIG. **1**;
(7) FIG. **6** is a perspective side view of another exemplary attachment apparatus, that can be utilized in the aircraft engine of FIG. **1** with a section exploded;
(8) FIG. **7** is a partial cross-section elevational view of the attachment apparatus of FIG. **6**;

(9) FIG. **8** is a bottom view showing an interior-facing surface of the backer plate and the partially installed attachment apparatus of FIG. **8**;
(10) FIG. **9** is a flow chart diagram of an exemplary process of attaching two components having different coefficients of thermal expansion.
(11) Elements in the figures are shown for simplicity and clarity and have not been drawn to scale. The dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present disclosure. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present disclosure. Certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required.
(12) The terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.
DETAILED DESCRIPTION
(13) The following description is not to be taken in a limiting sense, but is made merely for the purpose of describing the general principles of exemplary embodiments. Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.
(14) As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
(15) The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.
(16) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
(17) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. The approximating language may refer to being within a +/−1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values, and/or endpoints defining range(s) of values.
(18) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.
(19) Notably, the thermal expansion coefficients of the CMC exhaust liner and the typical metal exhaust duct do not closely match. As such, when exposed to the high temperature environment present in the exhaust nozzle of an aircraft engine, the metal duct expands more than the CMC exhaust liner, which can lead to undesirable results such as stress and displacements.
(20) Notably, the thermal expansion coefficients of the CMC exhaust liner and the typical metal exhaust duct do not closely match. As such, when exposed to the high temperature environment

present in the exhaust nozzle of an aircraft engine, the metal duct expands more than the CMC exhaust liner, which can lead to undesirable results such as stress and displacements.

(21) Conventional techniques for handling the high temperatures present in/around an aircraft engine include attaching a metal exhaust protective liner directly to the metallic/non-metallic aircraft component to be protected (e.g., metallic duct of an exhaust nozzle of the aircraft engine) using bolts and formed/machined hanger systems. The thermal growth of both metal components is quite similar, which allows both components to be connected using bolts without regard for the differential in their thermal growth characteristics. Other techniques include attaching a ceramic matrix composite (CMC), a polymer matrix composite (PMC) protective to the aircraft component to be protected, since the CMC/PMC materials are lighter and is capable of withstanding higher temperatures than the typical metallic protective liner.

(22) In the aviation industry, there is a desire for components that are made of lighter materials rather than conventional metal materials. Ceramics and their composites such as ceramic matrix composites (CMCs) provide a lightweight material option that is durable at various temperatures and thus desirable for incorporation into aircraft engines. The ceramic composite materials often need to be combined with and attached to aircraft engine components made of conventional metals. Such metal to non-metal attachments are sometimes used in high-temperature environments, for example, an exhaust nozzle of a turbofan aircraft engine.

(23) Since the ceramic matrix composite components and the conventional metals (e.g., titanium) have different thermal expansion coefficients, the present disclosure provides a solution for attaching components of different thermal expansion characteristics to each other for use in a high-temperature environment. The solution provides an attachment apparatus that securely attaches a CMC or the like non-metallic liner to a metallic component of an aircraft engine while preventing the head of the fastener that attaches the CMC liner to the metal duct from being pulled through the CMC liner as a result of thermal expansion of the CMC liner and/or the metallic component. As such, the embodiments of the attachment apparatus described herein provide an improved and prolonged attachment of the CMC liner to the metal and protect the metal duct from deterioration and/or failing even at the high temperatures present in aircraft engines.

(24) FIG. **1** illustrates an exemplary exhaust nozzle **10** of an exemplary aircraft engine **100**. The engine **100** may be a jet engine, for example, a turbofan engine. The exemplary exhaust nozzle **10** has a generally cylindrical shape with a first component or metal duct **12** enclosing an interior **14** of the exhaust nozzle **10**. In the embodiment illustrated in FIG. **1**, a second component or protective liner **20** is attached to an inwardly-facing or first surface **15** of the first component **12** using a plurality of hanger-like attachment apparatuses **30**, exemplary embodiments of which will be described hereinbelow.

(25) In the exemplary embodiment depicted in FIG. **1**, the second component **20** is shown as being attached to the first component **12** via eight attachment apparatuses **30** that are positioned along a portion of the second component **20** that is closer to the rear of the exhaust nozzle **10**. It will be appreciated that, depending on the needs of a specific installation and the size of the exhaust nozzle **10** and/or second component **20**, less than eight or more than eight attachment apparatuses **30** may be used. In addition, in some embodiments, the second component **20** may be attached to the first component **12** via two different types of attachment apparatuses, for example, the portion of the second component **20** that is closer to the rear of the exhaust nozzle **10** may be attached to the first component **12** via a suitable number (e.g., 4, 6, 8, 10, 12) of attachment apparatuses **30** as shown in FIG. **1**, and the portion of the second component **20** that is closer to the front of the exhaust nozzle **10** may be attached to the first component **12** via a suitable number (e.g., 4, 5, 8, 10, 12) of different attachment apparatuses, which are described in more detail in application Ser. No. 17/863,993 entitled SYSTEMS AND METHODS FOR ATTACHMENT OF MATERIALS HAVING DIFFERENT THERMAL EXPANSION COEFFICIENTS, filed Jul. 13, 2022, which is incorporated herein in its entirety.

(26) FIGS. **2-3** illustrate an embodiment of an attachment apparatus **30** (also referred to herein as "an attachment hanger" or just "hanger") for attaching a second component **20** to a first component **12**. It will be understood that the second component **20** and the first component **12** have different coefficients of thermal expansion. In non-limiting examples the second component **20** can be a CMC liner and the first component **12** can be a metal duct of an exhaust nozzle **10** of an aircraft engine. While reference is being made to attachment of a CMC liner **20** to the metal duct **12** of an exhaust nozzle **10**, it will be appreciated that the CMC liner **20** is just an exemplary material that may be used as a protective liner for the metal duct **12** of the exhaust nozzle **10**, and that any similar non-metallic material (e.g., polymer matrix composite (PMC) or the like) suitable for lining the interior of the metal duct **12** of the exhaust nozzle **10** by way of having suitable performance at elevated temperatures experienced in conventional or afterburning turbine exhausts for a high temperature environment such as an interior of an exhaust nozzle **10** of an aircraft engine may be used instead. By the same token, while reference is made to attachment of a protective liner **20** to a metal duct **12** of an exhaust nozzle **10** of an aircraft, it will be appreciated that the metal duct **12** of the exhaust nozzle **10** is just an exemplary first component **12** of an aircraft to which a second component **20** having a different coefficient of thermal expansion may be attached via the apparatus **30**, and that the apparatus **30** may be used to attach any two components that are made of materials having different coefficients of thermal expansion.
(27) It will be understood that components of the gas turbine engine such as the liner may comprise a composite material, such as a ceramic matrix composite (CMC) material, which has high temperature capability. As used herein, CMC refers to a class of materials that include a reinforcing material (e.g., reinforcing fibers) surrounded by a ceramic matrix phase. Generally, the reinforcing fibers provide structural integrity to the ceramic matrix. Some examples of matrix materials of CMCs can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), oxide ceramics (e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al.sub.2O.sub.3), silicon dioxide (SiO.sub.2), aluminosilicates, or mixtures thereof), or mixtures thereof. Optionally, ceramic particles (e.g., oxides of Si, Al, Zr, Y, and combinations thereof) and inorganic fillers (e.g., pyrophyllite, wollastonite, mica, talc, kyanite, and montmorillonite) may also be included within the CMC matrix.
(28) Some examples of reinforcing fibers of CMCs can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), non-oxide carbon-based materials (e.g., carbon), oxide ceramics (e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al2O3), silicon dioxide (SiO2), aluminosilicates such as mullite, or mixtures thereof), or mixtures thereof.
(29) Generally, particular CMCs may be referred to as their combination of type of fiber/type of matrix. For example, C/SiC for carbon-fiber-reinforced silicon carbide; SiC/SiC for silicon carbide-fiber-reinforced silicon carbide, SiC/SiN for silicon carbide fiber-reinforced silicon nitride; SiC/SiC—SiN for silicon carbide fiber-reinforced silicon carbide/silicon nitride matrix mixture, etc. In other examples, the CMCs may be comprised of a matrix and reinforcing fibers comprising oxide-based materials such as aluminum oxide (Al.sub.2O.sub.3), silicon dioxide (SiO2), aluminosilicates, and mixtures thereof. Aluminosilicates can include crystalline materials such as mullite (3Al2O3 2SiO2), as well as glassy aluminosilicates.
(30) In certain embodiments, the reinforcing fibers may be bundled and/or coated prior to inclusion within the matrix. For example, bundles of the fibers may be formed as a reinforced tape, such as a unidirectional reinforced tape. A plurality of the tapes may be laid up together to form a preform component. The bundles of fibers may be impregnated with a slurry composition prior to forming the preform or after formation of the preform. The preform may then undergo thermal processing, such as a cure or burn-out to yield a high char residue in the preform, and subsequent chemical processing, such as melt-infiltration with silicon, to arrive at a component formed of a CMC material having a desired chemical composition.

(31) Such materials, along with certain monolithic ceramics (i.e., ceramic materials without a reinforcing material), are particularly suitable for higher temperature applications. Additionally, these ceramic materials are lightweight compared to superalloys, yet can still provide strength and durability to the component made therefrom. Therefore, such materials are currently being considered for many gas turbine components used in higher temperature sections of gas turbine engines, such as airfoils (e.g., turbines, and vanes), combustors, shrouds and other like components, that would benefit from the lighter-weight and higher temperature capability these materials can offer.
(32) In the illustrated embodiment, the attachment apparatus **30** includes a mounting plate or mounting member **32** having an outwardly-facing or first surface **34** and an inwardly-facing surface or second surface **36** when mounted. The first surface **34** may be at least in part curved to complement the curvature of the first surface **15** of the first component **12**.
(33) In the embodiment shown in FIGS. **2-3**, the mounting member **32** is coupled relative to the first component **12** (which may be metallic or non-metallic) via a coupling member **38** (e.g., a track, a clip, or the like) attached to the first surface **15** of the first component **12**. In certain aspects, the track or clip **38** is generally C-shaped to define a channel **39** that receives at least a portion of the mounting member **32** and permits the mounting member **32** to slide axially in two directions (shown via two-directional arrows **17** in FIG. **1**) therein. As such, when the second component **20** is attached to the first component **12** via the attachment apparatus **30**, the slidable engagement of the mounting member **32** and the track or clip **38** attached (e.g., via welding) to the duct **12** provides some freedom of movement to the mounting member **32** within the channel **39** (in the directions indicated by the arrows **17** and **19** in FIGS. **1** and **2**), advantageously accommodating for possible thermal expansion of the first component **12** and/or the second component **20**. In addition, since the second component (e.g., a CMC or PMC-based material) is significantly stronger in-plane than it is through the thickness thereof, the use of the attachment apparatuses **30**, **130**, **230** in conjunction with the backer plates **60**, **160**, **260** as described herein significantly spreads out the through-thickness point load to significantly increase the ability of the second component **20** to withstand high through-thickness loads, thereby advantageously significantly reducing the chances of failure of the second component **20** even at high through-thickness loads. Furthermore, as a result of the unique configurations of the attachment apparatuses **30**, **130**, **230** described herein, the manufacture of the attachment apparatuses **30**, **130**, and **230** is advantageously simple, fast, and cost-effective.
(34) In the non-limiting example shown in FIGS. **2-3**, the attachment apparatus **30** further includes an attachment plate or member **40** opposite the mounting member **32** and a support post **48** interconnecting the mounting member **32** and the attachment member **40**. Notably, while the term “plate” is use in connection with the mounting member **32** and the attachment member **40**, the term “plate” as used herein may refer to any substantially flat structure or any other three-dimensional structure, and equivalents thereof, including those structures having one or more portions that are not substantially flat but curved along one or more axis. While the mounting member **32** and the attachment member **40** are interconnected by one support post **48** in the illustrated embodiment, it will be appreciated that, instead of one support post **48**, any suitable number of support posts can be utilized including but not limited to two or four support posts. In some embodiments, the mounting member **32**, the attachment member **40**, and one or more support posts **48** may form a unitary monolithic body.
(35) The attachment member **40** has an outwardly-facing or first surface **42** and an inwardly-facing or second surface **44**. The second surface **44** may be at least in part curved to complement the curvature of the outwardly-facing surface **21** of the second component **20**. The attachment member **40** further includes apertures **46** that extend through the thickness of the attachment member **40** from the first surface **42** and the second surface **44** to permit portions of fasteners **50** to pass through the attachment member **40**, as shown in FIGS. **2** and **3**.

(36) Given the possible thermal expansion of the first component **12** and/or the second component **20**, and since the second component **20**, which could be made, for example of CMC, PMC, or the like material, would not be as strong as a metallic material would be, to prevent the heads **52** of the fasteners **50** to be pulled outwardly through the thickness of the second component **20** as a result of mechanical loads induced in operation and differences in resulting thermal expansion, the attachment apparatus **30** illustrated in FIGS. **2** and **3** advantageously includes a backer plate **60** apertures **65** configured to permit a portion (e.g., the threaded shaft of the fastener **50**) to pass therethrough. In some embodiments, the backer plate **60** has an exterior profile that is contoured to match the needs of a particular installation of the attachment apparatus **30**

(37) As shown in FIG. **3**, with the fasteners **50** being fully tied down, the heads **52** of the fasteners **50** securely attach the outwardly-facing or second surface **61** of the backer plate **60** to the interior-facing surface **23** of the second component **20**. In addition, the shaft **54** of the fastener **50** (at least a portion of which is threaded) passes through the second component **20** and through the apertures **46** in the attachment member **40**, being attached relative to the attachment member **40** by nuts **56** (which may be self-locking nuts), thereby securing both the attachment member **40** and the backer plate **60** relative to second component **20**.

(38) As shown in FIGS. **2-3**, the nuts **56** may be tied onto thermal spacers **58** (through which the shaft **54** of the fastener **50** passes). Generally, the thermal spacers **58** may be made from alloys specifically selected to have a relatively high coefficient of thermal expansion to make up for the relatively low thermal expansion coefficient of the second component **20**. It will be appreciated that Belleville washers may be used instead of thermal spacers **58** in certain implementations. The use of the thermal spacers **58** may accommodate for possible thermal expansion of the second component **20** and/or the fasteners **50**, keeping the attachment of the backer plate **60** and the attachment member **40** to the second component **20** more secure.

(39) In the illustrated embodiment, the backer plate **60** has a countersunk configuration that includes a first recessed portion **62** and a second recessed portion **64**, as well as an inwardly-facing or first surface **63** of the backer plate **60**, and the aperture **65** of the backer plate is recessed relative to the first surface **63** to form a seat for the head **52** of the fastener **50** when the fastener **50** is secured to the backer plate **60**, while also securely attaching the backer plate **60** to the interior-facing surface **23** of the second component **20**.

(40) The backer plate **60** may be made of a metallic or non-metallic (e.g., ceramic matrix composite, polymer matrix composite, etc.) material, which may have the same or different coefficient of thermal expansion in comparison to the second component **20**. Generally, a metallic backer plate **60** would provide more forgiveness in tolerances as compared to a non-metallic (e.g., a ceramic matrix composite or a polymer matrix composite material, but would have a larger thermal growth mismatch with the second component **20** (which may be PMC or CMC) than a non-metallic (e.g., a PMC or CMC) backer plate **60**. Without wishing to be limited to theory, the use of the backer plate **60** to secure the attachment member **40** of the attachment apparatus **30** to the second component **20** spreads out the through-thickness point load to increase the ability of the second component **20** to withstand high through-thickness loads, and thereby restricts/prevents the heads **52** of the fasteners **50** from being pulled through the thickness of the second component **20** in an outwardly direction even as a result of high through-thickness loads that may be incurred by the second component **20** in operation (e.g., due at least in part to thermal expansion). As can be seen in FIG. **4**, the exemplary backer plate **60** may have a generally ellipsoid/oval exterior contour having curved and/or straight portions, but it will be appreciated that the backer plate **60** may be generally rectangular or circular in certain embodiments.

(41) With reference to FIGS. **3-4**, when the heads **52** of the fasteners **50** are at least in part seated within the aperture **65** of the backer plate **60**, the abutment of the heads **52** of the fasteners **50** with the first **63** of the backer plate **60**, and the abutment of the second **61** of the backer plate **60** with the interior-facing surface **23** of the second component **20** create a more even load distribution in

response to a primary load (the exemplary direction of which is shown by way of arrows **11** in FIGS. **3**, **5**, and **7**) and resistance to possible movement of the fasteners **50** in response to the forces of the primary load through the thickness of the second component **20**, providing point-load/buckling support and facilitating a more secure attachment of the attachment apparatus **30** to the second component **20**.

(42) As can be seen in FIG. **3**, the backer plate **60** is countersunk in such that heads **52** of the fasteners **50** are recessed relative to the first surface **63** of the backer plate **60** such that no portion of the heads **52** of the fasteners **50** protrudes inwardly relative to the interior-facing surface **23** of the second component **20**. The recessed arrangement of the heads **52** of the fasteners **50** within the aperture **65** of the backer plate **60** advantageously keeps the metallic heads **52** of the fasteners **50** from direct impingement of the high temperature exhaust gases that may be present within the exhaust, nozzle **10** or another compartment of an aircraft engine, thereby protecting the heads **52** from thermal expansion and/or deformation and causing less flow path disruption through the interior **14** of the exhaust nozzle **10**. In some embodiments, the overall low profile (i.e., low overall thickness) of the backer plate **60**, in combination with the use of the smaller-size fasteners **50** adapted for being countersunk within the aperture **65** of the backer plate **60** advantageously results in a backer plate design that is aerodynamic and provides a less observable electromagnetic effect.

(43) FIG. **5** illustrates another non-limiting example of an attachment apparatus **130** (also referred to herein as "an attachment hanger" or simply "hanger") for attaching a second component **20** (which may be made, for example, of CMC, PMC, or the like) to a (metallic or non-metallic) first component **12**. The attachment apparatus **130** is configured for attaching a protective material (e.g., a CMC liner, PMC liner, or the like) to a material intended to be protected (which may be a metallic/non-metallic duct of an exhaust nozzle of an aircraft engine, or another metallic/non-metallic aircraft component, or a metallic/non-metallic component of a non-aircraft vehicle) and has an overall construction that is generally similar to that of the attachment apparatus **30** described above, with some differences highlighted below. For ease of reference, structural aspects of the attachment apparatus **130** that are similar to aspects of the attachment apparatus **30** described above with reference to FIGS. **2-4** have been designated in FIG. **5** with similar reference numbers, but prefaced with a "1."

(44) Similarly to the backer plate **60**, the backer plate **160** includes an aperture **165** and may be made of a metallic or non-metallic (e.g., ceramic composite, polymer composite, or the like) material. Similarly to the backer plate **60**, the backer plate **160** restricts/prevents the heads **152** of the fasteners **150** from being pulled through the thickness of the second component **20** in an outwardly direction in response to a primary load (the exemplary direction of which is shown by way of arrows **11** in FIG. **5**, and which may occur, e.g., as a result of thermal expansion).

(45) Unlike the backer plate **60**, the backer plate **160** does not-include recessed portions akin to the recessed portions **62**, **64** of the backer plate **60**. Instead, the backer plate **160** may be substantially planar having a first surface **163** and a second surface **161**, opposite the first surface. It will be understood that the first surface **163** and the second surface **161** can be parallel to each other along their lengths. The backer plate **160** can be considered generally planar even if a first surface **163** may be at least in part curved to complement the curvature of the interior-facing surface **23** of the second component **20**. Without wishing to be limited to theory, the backer plate **160** is less sensitive to thermal expansion-induced distress and possible misalignment of the fasteners **50** than the backer plate **60**.

(46) As can be seen in FIG. **5**, the fasteners **150** have a protruding configuration, where the heads **152** of the fasteners **150** protrude beyond the backer plate **160** and into the flow path within the interior **14** of the exhaust nozzle **10**. To that end, the fasteners **150** may be different from the fasteners **50** described above. For example, in the embodiment shown in FIG. **5**, the head **152** of each fastener **150** includes a narrower portion **153** and an expanded portion **155**, with the expanded portion **155** having a larger diameter than the narrower portion **153** and providing a more

secure/larger surface area attachment of the head **152** of the fastener **150** to the inwardly-facing surface of the second component **20**.
(47) FIGS. **2-5** above depict non-limiting examples of attachment hangers **30**, **130** that are configured to be secured to the second component **20** via two fasteners **50**, **150**. FIG. **6** illustrates a non-limiting exemplary attachment apparatus **230** utilizing one fastener **250**. Again, for ease of reference, structural aspects of the attachment apparatus **230** that are similar to aspects of the attachment apparatus **30** described above with reference to FIGS. **2-4** have been designated in FIG. **5** with similar reference numbers, but prefaced with a "2."
(48) In the illustrated embodiment, the attachment apparatus **230** includes a mounting member **232**. The mounting member **232** may be considered a contoured plate. As illustrated, the mounting member **232** includes a first shoulder **235** and a second shoulder **237**, with an opening **241** located within a generally centralized portion of the mounting member. In the embodiment shown in FIG. **7**, the shoulders **235**, **237** of the mounting member **232** are coupled relative to the (metallic/non-metallic) first component **12** via a coupling member **238** (e.g., a track, a clip, or the like) attached (e.g., via welding or the like) to the inwardly-facing surface **15** of the first component **12**.
(49) In certain aspects, the track or clip **238** is generally C-shaped to define a channel **239** that receives the shoulders **235**, **237** of the mounting member **232** and permits the shoulders **235**, **237** of the mounting member **232** to slide axially in two directions (which are illustrated by arrows **17** in FIG. **1**) therein. As such, when the second component **20** is attached to the first component **12** via the attachment apparatus **230**, the slidable engagement of the mounting member **232** and the track or clip **238** attached to the first component **12** provides some freedom of movement to the mounting member **232** (in the directions indicated by the double arrow **219** in FIG. **6**) within the channel **239**, advantageously accommodating for possible thermal expansion of the first component **12** and/or the second component **20**.
(50) In the embodiment shown in FIGS. **6-7**, the attachment apparatus **230** further includes an attachment member **240** opposite the mounting member **232**. It will be understood that a plurality of support posts **248** are included in the attachment apparatus. In the illustrated example, at least two support posts are shown. Each of the support posts **248** includes an opening or space **243**, which may effectively reduce an overall weight of the support posts **248** and the attachment apparatus **230**. It will be appreciated that any number of support posts **248** may be used.
(51) The attachment member **240** has an outwardly-facing or first surface **242** and an inwardly-facing or second surface **244**. The attachment member **240** further includes an opening **246** that extends through the thickness of the attachment member **240** from the first surface **242** and the second surface **244** to permit the shaft **254** of the fastener **250** to pass through the attachment member **240**, as shown in FIGS. **7-8**. In some embodiments, the attachment member **240** may include two flange-like self-alignment members **247**, **249** extending from the second surface **244**. In the embodiment shown in FIG. **7**, the self-alignment members **247**, **249** are located between the attachment member **240** of the attachment apparatus **230** and a recessed portion **262** of the backer plate **260**, providing room for the thermal spacer **258** and providing a mechanism for ensuring proper alignment between the attachment apparatus **230** and/or the backer plate **260** during installation.
(52) Given the possible thermal expansion of the first component **12** and/or the second component **20**, and given that the second component **20** made of CMC/PMC material would not be as not as strong in the through-thickness direction as a second component that is metallic would be, to prevent the head **252** of the fastener **250** to be pulled in response to the primary load (shown by arrows **11** in FIG. **7**) outwardly through the thickness of the second component **20** (e.g., as a result of thermal expansion), the attachment apparatus **230** illustrated in FIGS. **7** and **8** advantageously includes a backer plate **260** including an aperture **265** configured, to permit a portion (e.g., the threaded shaft **254** of the fastener **250**) to pass therethrough.
(53) As shown in FIG. **6-8**, when assembled, the head **252** of the fastener **250** securely attaches the

outwardly-facing or first surface **261** of the backer plate **260** to the interior-facing surface **23** of the second component **20**. In addition, the shaft **254** of the fastener **250** (at least a portion of which is threaded) passes through the second component **20** and through the opening **246** in the attachment member **240**, being attached relative to the attachment member **240** by a nut **256** (which may be a self-locking nut), which secures both the attachment member **240** and the backer plate **260** relative to the second component **20**.

(54) As shown in FIGS. **6-7**, the nut **256** may be tied onto the first surface **242** of the attachment member **240**, with the second surface **244** of the attachment member **240** being in turn tied down (by the force exerted by the tying down of the nut **256**) onto a thermal spacer **258** (through which the shaft **254** of the fastener **250** passes). As mentioned above, the thermal spacer **258** may be made from alloys specifically selected to have a relatively high thermal expansion coefficient to make up for the relatively low thermal expansion coefficient of the second component **20**. In other words, in the configuration of FIG. **7**, the thermal spacer **258** is located between the second surface **244** of the attachment member **240** and the outwardly-facing surface **21** of the second component **20**. It will be appreciated that a Belleville washer may be used instead of the thermal spacer **258** in certain implementations. The use of the thermal spacer **258** may accommodate for possible thermal expansion of the second component **20** and/or the fastener **50**, keeping the attachment of the backer plate **260** and the attachment member **240** to the protective liner **20** more secure.

(55) In the illustrated embodiment, the backer plate **260** has a countersunk configuration that includes a recessed portion **262**, which is recessed (in an outwardly direction) relative to the inwardly-facing or first surface **263** (which is opposite the outwardly-facing or second surface) of the backer plate **260**, forming a seat for the head **252** of the fastener **250** when the fastener **250** is secured to the backer plate **260**, thereby securely attaching the backer plate **260** to the interior-facing surface **23** of the protective liner **20**.

(56) The backer plate **260** may be made of a metallic or non-metallic (e.g., ceramic composite, polymer composite, or the like) material. The backer plate **260** of FIG. **7** has a different exterior contour as compared to the exterior contour of the backer plate **160** of FIG. **5**. In addition, while no portion of the backer plate **160** is recessed into and passes through the aperture in the second component **20**, a portion of the backer plate **260**, namely, the recessed portion **262**, passes through a portion of the aperture in the second component **20** (notably, portions of the head **252** and the shaft **254** of the fastener **250** also pass through a portion of this aperture in the second component **20**). Without wishing to be limited to theory, the use of the backer plate **260** to secure the attachment member **240** of the hanger **230** to the protective liner **20** restricts/prevents the head **252** of the fasteners **250** from being pulled through the thickness of the second component **20** in response to the primary load (shown by arrows **11** in FIG. **7**) in an outwardly direction (e.g., as a result of thermal expansion). As can be seen in FIG. **8**, the backer plate **260** may have a generally circular exterior contour, but it will be appreciated that the backer plate **260** may be generally rectangular or ellipsoid/oval (with both curved and straight portions) in certain embodiments. As pointed out above, it will be appreciated that the backer plate **260** may be deliberately shaped in various configurations to increase the load bearing area or to decrease the load-bearing area, depending on the needs of a particular installation.

(57) With reference to FIGS. **7-8**, when the head **252** of the fastener **250** is seated within the recessed portion **262** of the backer plate **260**, the abutment of the head **252** of the fastener **250** with the first surface **263** of the backer plate **260**, and the abutment of the second surface **261** of the backer plate **260** with the interior-facing surface **23** of the second component **20** create a more even load distribution and resistance to possible movement of the shaft **254** and head **252** of the fastener **250** in response to the primary load (shown by arrows **11** in FIG. **7**) in an outwardly direction through the second component **20**, thereby providing point-load/buckling support and facilitating a more secure attachment of the attachment apparatus **230** to the second component **20**.

(58) As can be seen in FIG. **7**, the backer plate **260** is countersunk in that head **252** of the fastener

250 is recessed relative to the first surface **263** of the backer plate **260** such that no portion of the head **252** of the fastener **250** protrudes inwardly relative to the first surface **23** of the second component **20**, making the attachment apparatus **230** more aerodynamic overall. In addition, the recessed arrangement of the head **252** of the fastener **250** within the recessed portion **262** of the backer plate **260** advantageously keeps the metallic head **252** of the fastener **250** protected from direct impingement of the high temperature exhaust gases passing through the interior **14** of the exhaust nozzle **10**, thereby advantageously preventing possible deformation of the head **252** and/or causing less flow path disruption through the interior **14** of the exhaust nozzle **10**. Alternatively, the backer plate **260** may be a non-countersunk configuration (akin to the configuration of the backer plate **160** of FIG. **5**), where the backer plate **260** does not include a recessed portions akin to the recessed portion **262**, but instead has planar, non-recessed straight or curved first and second surfaces **261** and **263**.

(59) With reference to FIG. **9**, an exemplary method **300** of attaching a second component (e.g., a protective material) to a first component (e.g., metallic/non-metallic aircraft component or a non-aircraft component), where the first and second components have different coefficients of thermal expansion will now be described. For exemplary purposes, the method **300** is described in the context of attaching the attachment apparatus **30** described with reference to FIGS. **2-4** to the second component **20** and to the first component **12**, thereby attaching the second component **20** to the first component **12**, but it will be understood that embodiments of the method **300** may be implemented to attach various other embodiments of the attachment apparatuses **130** (FIG. **5**) and **230** (FIGS. **6-8**) to the first component and the second component **20**.

(60) In the illustrated embodiment, the method **300** of attaching two components having different coefficients of thermal expansion includes movably coupling a mounting member **32** of the attachment apparatus **30** to a first component **12**, the first component **12** having a first coefficient of thermal expansion (step **310**). As pointed out above, a track or clip **38** mounted to the first component **12** may include a channel **39** that receives at least a portion of the mounting member **32** and permits the mounting member **32** to slide axially therein in two directions (shown via two-directional arrows **17** in FIG. **1**).

(61) The method further includes mounting an attachment member **40** of the attachment apparatus **30** to a second component **20** having a second coefficient of thermal expansion (step **320**). As pointed out above, the attachment member **40** includes one or more apertures **46** therein that extend through the full thickness of the attachment member **40** and permit a shaft **54** of a fastener **50** to pass through the attachment member **40**.

(62) The method **300** further includes passing a shaft **54** of at least one fastener **50** through at least one aperture **65** of the backer plate **60**, through the second component **20**, and through at least aperture **46** of the attachment member **40** (step **330**). The apertures **65** of the backer plate **60** extend through the full thickness of the backer plate **60** to permit the shaft **54** of a fastener **50** to pass through the backer plate **60**. As pointed out above, depending on the intended application, the backer plate **60** may be deliberately shaped in various shapes and sizes to both increase the load bearing area (i.e., when the backer plate **60** is contoured to be of a maximum overall size suitable for a given installation) and reduce the electro-magnetic signature of the backer plate **60** (i.e., when the backer plate **60** is contoured to be of a minimum overall size suitable for a given installation).

(63) As described in more detail above, in some embodiments, the backer plate **60** may have a countersunk configuration including first and second recessed portions **62**, **64** that are recessed relative to the interior-facing surface **23** of the second component **20** and define seats for the heads **52** of the fasteners **50**, and enable the heads **52** of the fasteners to be recessed relative to the inwardly-facing surface **63** of the backer plate **60**, such that no portion of the heads **52** of the fasteners **50** protrudes inwardly relative to the interior-facing surface **23** of the second component **20**. Alternatively, the backer plate **160** may be constructed in a non-countersunk configuration, where the backer plate **160** does not include recessed portions that extend into the inwardly-facing

surface **23** of the second component **20** akin to the recessed portions **62**, **64** of the backer plate **60**. In some embodiments, the attachment apparatus **30**, **130** may be configured such that the backer plate **60**, **160** and the attachment member **40**, **140** of the attachment apparatus **30**, **130** are attached to the second component **20** via two fasteners **50**, **150**, while in other embodiments, the attachment apparatus **230** may be configured such that the backer plate **260** and the attachment member **240** of the attachment apparatus **230** are attached to the second component **20** via one fastener **250**.
(64) Without wishing to be limited to theory, the embodiments of the attachment apparatus described herein are easy and cost-effective to manufacture and easy to install, and provide more even load distribution of force exerted by the fastener(s) onto the second component **20** as a result of mechanical loads and thermal expansion that may occur during operation of an engine (e.g., an aircraft engine), thereby advantageously providing buckling support and resistance to possible movement of the fasteners **50** in an outwardly direction through the second component **20**, thereby facilitating a more secure and longer-life-cycle attachment of the attachment apparatuses **30** to the second component **20** and, in turn, a more secure and longer-life-cycle attachment of the second component **20** (e.g., CMC liner, PMC liner, or the like) to the first component **12** (e.g., metal or metal alloy duct of an exhaust nozzle **10** of an engine).
(65) Further aspects of disclosure are provided by the subject matter of the following clauses:
(66) There is provided an attachment apparatus, comprising: a mounting member configured to be movably coupled relative to a first component having a first coefficient of thermal expansion; an attachment member mountable to a second component having a second coefficient of thermal expansion, the attachment member including at least one aperture therein; at least one support post interconnecting the mounting member and the attachment member; a backer plate including at least one aperture having a first diameter; and at least one fastener having a shaft and a head. When assembled, the shaft passes through the at least one aperture of the backer plate, through the second component, and through the at least one aperture of the attachment member to attach the second component to the first component. The head of the at least one fastener has a second diameter that is larger than the first diameter of the at least one aperture of the backer plate.
(67) The apparatus may include a track or clip attached to the first component and configured to define a channel that permits the mounting member to slidably move therein in two directions.
(68) The mounting member, the attachment member, and the at least one support post of the attachment apparatus may be a single unitary monolithic body. At least one of the mounting member or the attachment member may be a plate. The plate may be a contoured plate including an aperture therein.
(69) The at least one post of the apparatus may be a singular post. At least a portion of the at least one post may include a cutout to reduce a total weight of the at least one post.
(70) When the apparatus is in use, a primary load direction may be perpendicular to the mounting member, the attachment member, and the backer plate.
(71) The backer plate may be contoured. The backer plate may have a first surface and a second surface, the second surface opposite the first surface and wherein the at least one aperture is recessed from the first surface. When the apparatus is assembled, the head of the at least one fastener may be flush with the first surface of the backer plate. Alternatively, when the apparatus is assembled, the head of the at least one fastener may protrude inwardly relative to the first surface of the backer plate.
(72) The second component may include an aperture configured to permit the at least one fastener to pass therethrough. A portion of the backer plate may pass through at least a portion of the aperture of the second component.
(73) The backer plate may be made of a material having a third coefficient of thermal expansion that is different from the second coefficient of thermal expansion.
(74) The at least one fastener may be two fasteners, and the at least one aperture of the backer plate may include two apertures to permit the two fasteners to pass therethrough and pass through the

second component and through the attachment member.
(75) The apparatus may further include at least one nut configured to engage the shaft of the at least one fastener and secure the at least one fastener relative to the attachment member; and at least one thermal spacer secured relative to the attachment member. The at least one thermal spacer may be positioned between the at least one nut and the attachment member. The attachment member may be positioned between the at least one nut and the at least one thermal spacer and the at least one thermal spacer may be positioned between the attachment member and the second component.
(76) A system for attaching the second component to the first component is also provided. In the system, the first component is an aircraft engine component and the second component is a protective liner configured to protect the first component. The system includes a plurality of attachment apparatuses identical to the attachment apparatus of claim **1**, and the attachment apparatuses are spaced from one another and movably coupled relative to the aircraft engine component and mounted to the protective liner to attach the protective liner to an inwardly-facing surface of the aircraft engine component such that the protective liner fully covers the inwardly-facing surface of the aircraft engine component. The first component of the system may be a metallic material and the second component of the system may be a ceramic material.
(77) Those skilled in the art will recognize that a wide variety of other modifications, alterations, and combinations can also be made with respect to the above-described embodiments without departing from the scope of the invention, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept.

Claims

1. An attachment apparatus, comprising: a mounting member movably coupled relative to a first component having a first coefficient of thermal expansion; an attachment member mounted to a second component having a second coefficient of thermal expansion, the attachment member including at least one aperture therein; at least one support post interconnecting the mounting member and the attachment member; a backer plate including at least one aperture having a first diameter; at least one fastener having a shaft and a head; and at least one nut threadably engaging threads of the shaft of the at least one fastener and securing the at least one fastener relative to the attachment member; wherein the shaft passes through the at least one aperture of the backer plate, through the second component, and through the at least one aperture of the attachment member to attach the second component to the first component; and wherein the head of the at least one fastener has a second diameter that is larger than the first diameter of the at least one aperture of the backer plate.
2. The apparatus of claim 1, further comprising a track or clip attached to the first component and configured to define a channel that permits the mounting member to slidably move therein in two directions.
3. The apparatus of claim 1, wherein the mounting member, the attachment member, and the at least one support post are a single unitary monolithic body.
4. The apparatus of claim 1, wherein at least one of the mounting member or the attachment member is a plate.
5. The apparatus of claim 4, wherein the plate is a contoured plate including an aperture therein.
6. The apparatus of claim 1, wherein at least a portion of the at least one support post includes a cutout to reduce a total weight of the at least one support post.
7. The apparatus of claim 1, wherein, when the apparatus is in use, a primary load direction is perpendicular to the mounting member, the attachment member, and the backer plate.
8. The apparatus of claim 1, wherein the backer plate is contoured.
9. The apparatus of claim 8, wherein the backer plate has a first surface and a second surface, the second surface opposite the first surface and wherein the at least one aperture is recessed from the

first surface.
10. The apparatus of claim 9, wherein the head of the at least one fastener is flush with the first surface of the backer plate.
11. The apparatus of claim 1, wherein the second component includes an aperture configured to permit the at least one fastener to pass therethrough, and wherein a portion of the backer plate passes through at least a portion of the aperture of the second component.
12. The apparatus of claim 1, wherein the backer plate is made of a material having a third coefficient of thermal expansion that is different from the second coefficient of thermal expansion.
13. The apparatus of claim 1, further comprising: at least one thermal spacer secured relative to the attachment member.
14. The apparatus of claim 13, wherein the at least one thermal spacer is positioned between the at least one nut and the attachment member.
15. The apparatus of claim 13, wherein the attachment member is positioned between the at least one nut and the at least one thermal spacer and the at least one thermal spacer is positioned between the attachment member and the second component.
16. A system for attaching the second component to the first component, wherein the first component is an aircraft engine component, and wherein the second component is a protective liner configured to protect the first component, the system comprising a plurality of attachment apparatuses identical to the attachment apparatus of claim 1, wherein the attachment apparatuses are spaced from one another and movably coupled relative to the aircraft engine component and mounted to the protective liner to attach the protective liner to an inwardly-facing surface of the aircraft engine component such that the protective liner fully covers the inwardly-facing surface of the aircraft engine component.
17. The system of claim 16, wherein the first component is a metallic material and the second component is a ceramic material.